REP-031 · Fruit Market: game concept and mechanics (Tabletop edition, rules version 4)

Game	Fruit Market. Design: Murshidi bin Md Hashim
Report date	2026-10-27 13:46 (Asia/Kuala_Lumpur)
Type	Final mechanics (worked out and lab-tested) and a concept summary for benchmarking against board games
Plan	PLAN-014, approved: "approve all the idea, work out the mechanics, and summarise to me game back highlighting the concept and game mechanics as i want to benchmark against similar genre board games and traditional popular board games"
Status	Design final for the build. The app still runs rules version 3 until PLAN-014 is built

Labels. [Sim] = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, to be checked in the paper playtest.

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1. The game on one page

Pitch. You run a fruit stall in a busy harbour market. Each day you buy fruit from the farmers' lorry, guess what the crowd will want and what your rivals will charge, then post your prices in secret. The cheapest stall makes the sale. But fruit goes off every night, so stock you can't sell is money rotting on your shelf. After 8, 10 or 14 days, the richest stall wins.

Players	2–4 (solo against computer shops in the app)
Length	Quick 8, Standard 10 or Long 14 days. App: about 15–20 minutes. Table: about 45–60 minutes **[Hyp]**
Age / weight	10+. Light-to-medium strategy **[Hyp]**: simple turns, with depth in reading the market and rivals
Win condition	Most cash at the end. Unsold fruit and upgrades are worth nothing. Ties go to the shop that sold more fruit
Elimination	None. Everyone plays every day
Luck vs skill	Luck from the order of the farmer, customer and event decks, softened by a one-day look ahead. Skill in buying, timing, pricing and reading rivals. [Sim] Hard computer beats Normal 62%

The four pillars (your brief).

Pillar	How the game delivers it
A real market	Supply comes off a lorry; demand comes from customers with budgets. One posted price per fruit; the cheapest stall sells first. The market pulse moves prices up after a shortage and down after a glut
Inventory and decay	Every fruit ages one step a night and sells for less as it ages; the oldest spoils. Shelves are limited. Fruits rot at different speeds

Pillar	How the game delivers it
Demand forecasting	The crowd queues a day ahead: the expected customer is face up, and the walk-ins are face down with their wanted fruits printed on the back. Tomorrow's event and tomorrow's lorry are also shown a day early
Strategy and undercutting	Sealed simultaneous pricing creates the "do I go $1 lower?" duel. Eight different plans are viable (section 4)

2. Components (tabletop)

Component	Count	Purpose
Fruit tokens	4 kinds (about 40 each)	Blueberry, watermelon, apple, mango
Market board	1	Farmers market (crates by arrival day), customer queue, market pulse track, day track
Shop mats	4	Shelf (8 spaces), crates by arrival day, "for sale" crate
Price boards and screens	4	One slider per fruit ($1–$14), hidden until the reveal
Farmer cards	20	Tomorrow's lorry: 3 fruits in quantity plus 1 of the fourth; some mark one fruit as arriving green
Customer cards	24	Front: name, wanted fruits, quantity, top price for fresh fruit. Back: fruit icons only
Event cards	13	One a day from Day 2, each announced a day early
Favour cards	12 (6 kinds × 2)	One-off bonuses earned by serving a whole order alone
Upgrade cards	6	Evening draft
Trait cards	6	Pick 1 of 3 at the start
Coins, cubes, markers		Cash; one market pulse cube per fruit; day marker

Crates instead of sliding. Fruit sits in a crate marked with the day it arrived. Its age (today minus that day) gives its freshness from the chart on the mat. Nothing needs moving each night, and the app shows the same thing as a decay track.

3. A day, step by step

1. **Morning**
    - Reveal today's event (announced yesterday) and turn up tomorrow's.
    - **The lorry arrives:** put today's farmer card's fruit into the farmers market. Green fruit goes to the green crate.
    - Turn up **tomorrow's lorry** card.
2. **Buying**
    - **Poorest shop first** (ties go round the table).

    - Each shop buys up to **4 fruits** (5 with a Handcart), paying the cost for that freshness.
    - Shelf limit 8 (14 with a Bigger Shelf).
3. **The crowd**
    - Customers still waiting from yesterday stand at the front.
    - Today's expected customer and today's walk-ins are turned face up.
    - Draw **tomorrow's queue**: 1 expected customer face up, and the walk-ins face down (backs show their fruits).
4. **Pricing (secret)**
    - Behind your screen, set **one price per fruit** on your price board.
    - Put the fruit you are willing to sell in your "for sale" crate.
    - Everyone reveals at once.
5. **Sales**
    - Customers are served in queue order. For each fruit a customer wants, they buy from the **cheapest stall**.
    - Your price must not exceed what they pay for your fruit's freshness: their top price + market pulse + today's event − the age drop.
    - You hand over your oldest fruit that is fresh enough.
    - **Ties:** a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.
    - Paid on the spot.
    - Serve a customer's whole order by yourself to draw a **favour card** (one a day).
    - An unserved new customer waits until tomorrow; a waiting customer then leaves.
6. **Market pulse** (for each fruit)
    - **Shortage:** a customer still wants it, so the cube goes up $1 (to +$2 at most).
    - **Glut:** at least players + 1 ripe ones are left at the farmers market, so the cube goes down $1 (to −$1 at least).
    - Otherwise the cube moves one step back towards 0.
7. **Evening**
    - Upgrade draft: each shop sees 3 upgrades and may build any it can afford.
    - Evening favour cards can be played.
8. **Night**
    - All fruit ages one step, and fruit past its last step spoils.
    - The Juice Press turns up to 3 spoiling fruits into $2 each.
    - The Cool Room ages your fruit only every other night.
    - Night events apply (Health Inspector fines, Heatwave ageing).

Supply and demand on a table (no formulas).

- **Lorry:** the card's three fruits arrive at the printed number (2–4); the fourth fruit arrives as 1.
    - 2 players: one less of each listed fruit (minimum 1).
    - 4 players: one more of every fruit.
- **Walk-ins:** players − 2 in the first half of the game, players − 1 in the second half.
    - 2 players: 0 → 1.
    - 3 players: 1 → 2.
    - 4 players: 2 → 3.
- [Sim] These printed rules reproduce the old formula flow almost exactly. In 3 players, about 10 fruits a day are delivered, and customers want about 9 units a day early and 13 late.

Optional rule: Market Days (off by default, for the paper playtest).

- Every third day is a fair: +1 walk-in, and **no lorry**.
- Tomorrow's lorry shows "Market Day", so shops must stock up the day before.
- See section 6 for why it is optional.

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4. Fruit: price, decay and playstyle

	🫐 Blueberry	🍉 Watermelon	🍎 Apple	🥭 Mango
Playstyle	Quick flipper	Steady bulk seller	The keeper	Speculator
Lorry cost fresh / okay / not so / green	$2 / $1 / $1 / –	$3 / $2 / $1 / $1	$3 / $2 / $1 / –	$4 / $2 / $1 / $3
Customer top price (fresh)	$4–7	$6–9	$5–9	$7–11
Price drop by age, oldest → freshest	−4 · −2 · 0	−3 · −1 · −1 · 0 · 0	−2 · −1 · −1 · −1 · 0 · 0 · 0	−3 · −1 · 0 · 0
Shelf life	3 nights	5 nights (+1 green)	7 nights	4 nights (+2 green)
Risk	High: steepest decay	Medium	Lowest	Highest: wait to ripen, fast spoil, few delivered
[Sim] Net profit per fruit bought	$3.51	$3.15	$3.00	$5.15
Best moment	Sell the day you buy	Big orders, Heatwave	Stock for a Lorry Breakdown or shortage	Buy green before a Mango Craze or a shortage

The more a fruit can hurt you, the more it pays.

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5. Events, traits, upgrades and favours

Events (13 cards; one a day from Day 2, each shown a day early):

Event	Effect
Festival	Customers pay +$3, plus an extra walk-in
Blueberry Glut *(new, replaces Slow Season)*	+1 blueberry per player delivered; blueberry customers pay −$1
Heatwave	Watermelons +$4; all fruit ages an extra step tonight
Apple Blight *(new, replaces Rainy Day)*	No apples delivered; apple customers pay +$1
Mango Craze	Mangoes +$3; the Influencer wants 3
Superfood Trend	Blueberries +$3; the Gym Club wants 4
Good Harvest *(scaled)*	+1 of every fruit per 2 players (rounded up); lorry prices −$1

Event	Effect
Bad Harvest	Half delivery (rounded up); customers pay +$2
Price Hike	Lorry prices +$1
Minimum Price	Price floors per fruit
Health Inspector	$2 fine per not-so fruit tonight. Never the final day's event (draw again)
Lorry Breakdown	No delivery today
Hotel Order	The Hotel Chef orders 3 apples, 2 watermelons and 2 blueberries

Traits (pick 1 of 3 at the start, public):

Trait	Effect
Haggler	Wins ties against richer shops, even over fresher fruit
Big Basket	+1 fruit a day
Bargain Hunter	First fruit each day $1 cheaper
Charmer	Start with 3 favour cards; hold 4
Local Favourite	+$4 per customer you serve completely alone
Nest Egg	Start with $15 instead of $10

[Sim] 31–38% wins each in 3 players.

Upgrades (evening draft, 3 offered, built once):

Upgrade	Cost	Effect
Handcart	$6	Buy 5 a day
Signboard	$8	+$1 per fruit sold
Cool Room	$7	Ages every other night; Heatwave-proof
Bigger Shelf	$3	14 spaces
Juice Press	$4	+$1 per older fruit sold, and up to 3 spoiling fruits a night become $2 each
Shop Bell	$4	Draw 2 favour cards; hold 4

Favour cards (2 of each): Free Cash ($3), Free Fruit, Good Riddance (sell ripe stock back at $1 each), Fridge (all fruit one step fresher), Favoured Trade (win ties today), Priority (buy first tomorrow).

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6. Balance evidence (computer players)

Measure	Rules 3 (app now)	Rules 4 (this design)
Win share of 8 plans, 3 players (fair 33%)	23–37%	25–42% (Ripen at the top)
Win share, 4 players (fair 25%)	20–28%	18–25%

Measure	Rules 3 (app now)	Rules 4 (this design)
Hard beats Normal (2 players)	60%	56–62%
Comeback from last at half-time (3p / 4p)	26% / 19%	28–32% / 18–19%
Price-war sales at $2 or less (3p)	6%	5%
Quick (8 days): 4-plan table	–	19 / 30 / 30 / 21 (even)
Long (14 days): 4-plan table	Reserve 42%	Ripen 49% (watch item)
Traits (3 players)	30–37%	31–38%

Market Days (optional rule):

- Every third day with +1 walk-in gives the most even plans of anything tested (24–35%).
- It also gives 38% comebacks and spoilage halves, because shops stock ahead.
- But Hard and Normal computers become even (50%), because the computers don't plan ahead.
- [Hyp] People who plan ahead may make it the most skilful mode. The paper playtest decides.

Open items for the build:

- Shop Bell, Juice Press and Cool Room are worth little on their own under posted prices (12.8–14.5%, against 13.5% for no upgrade). They need retuning.
- Long games favour the ripening plan.
- The computer players need proper posted-price logic.

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7. For your benchmark: mechanism profile

Mechanism names follow the usual board-game catalogue terms, so you can compare like for like.

Mechanism	In Fruit Market	Weight in this game
Simultaneous action selection / sealed pricing	Secret price board, revealed together	Core
Lowest-price-wins competition (a reverse auction)	Cheapest stall serves the customer	Core
Dynamic market pricing	Market pulse after shortages and gluts; lorry prices by freshness	Core
Perishable inventory / resource decay	Fruit ages nightly, loses value, spoils	Core
Demand forecasting / hidden information	Face-down queue with fruit-icon backs; tomorrow's lorry and event shown	Core
Open market buying	Shared farmers market, first come, first served	Medium
Catch-up turn order	Poorest buys first	Medium
Variable player powers	Traits	Light
Card drafting	Evening upgrade draft (3 offered)	Light

Mechanism	In Fruit Market	Weight in this game
Event deck with look ahead	One event a day, announced a day early	Medium
Contract fulfilment	Serve a whole order alone for a favour	Light
Engine building	Upgrades that pay each day	Light

Starting points for comparison (well-known games sharing a mechanism; please verify details in your own benchmark):

Game	Shared idea	Where Fruit Market differs
Power Grid	A resource market whose prices rise as supply runs out; the trailing player buys first	Fruit also rots, and selling prices are set by the players in secret
Food Chain Magnate	Customers buy from the cheapest restaurant; price-setting competition	Much lighter; demand is forecast from a visible queue rather than built by marketing
For Sale / Modern Art	Blind or sealed bidding duels	Bidding is for customers each day, and goods spoil
Monopoly (traditional)	Win by cash; rents and luck	No elimination, no property; every day is a fresh price duel; catch-up turn order
Catan (traditional)	Variable supply each turn	Supply is shown a day ahead and bought, not rolled; no trading between players
Sushi Go / 7 Wonders	Simultaneous play and drafting	Drafting is a side system (upgrades); the core is pricing and stock

Possible distinctive hooks for your benchmark to confirm: perishable stock with a different decay speed per good; one posted price per good settled by a simultaneous reveal; and a self-adjusting price track driven by unmet demand.

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8. Next steps

1. **Paper playtest:**
    - print the cards with these numbers;
    - 3 sessions of 3–4 players, with Market Days on in one;
    - time each day.
2. **Build PLAN-014 in the app** (rules version 4):
    - new pricing screen (four sliders);
    - market pulse on the board;
    - queue silhouettes;
    - printed supply and demand;
    - event changes;
    - Market Days as an option in the game set-up;
    - computer posted pricing;
    - upgrade retune.
3. **Benchmark** using section 7, then feed changes back before the build is finalised.

Lab files (sweeps 16–18) are archived in `docs/workflow/reports/img/REP-031-lab.tar.gz`.